

The standard deviation controls the width. In the left plot, a standard deviation of 10 gives a tall, narrow spike, and a standard deviation of 100 gives a low, flat smear. The peak gets lower as the curve gets wider because every curve still contains the same 10,000 values, so spreading them out lowers the height. The mean has no effect on shape. In the right plot, each curve has the same shape and just slides left or right.

Here's what I measured from each sample:

std you asked for	measured std	measured variance	within ± 1 std	within ± 2 std	range of values
10	9.88	97.5	69.0%	95.7%	63 to 138
25	24.83	616.3	68.9%	95.5%	-11 to 190
50	49.86	2,486.3	68.5%	95.5%	-133 to 292
100	100.87	10,174.9	68.5%	95.2%	-299 to 480

Three things stand out in these results.

1. **The measured standard deviation almost matches the one you asked for.** It's slightly off because it comes from a random sample, not the ideal curve. More samples would bring it closer.
2. **Variance is the standard deviation squared.** For example, $9.88^2 \approx 97.5$ and $49.86^2 \approx 2,486$. That's why variance grows so fast: doubling the standard deviation multiplies the variance by four. Variance is the average squared distance from the mean, and the standard deviation is its square root. People usually talk about the standard deviation because it's in the same units as the data (dollars, not "squared dollars"), so you can read it directly off the x-axis.
3. **The percentages stay the same at every width.** About 68% of the values land within one standard deviation of the mean, and about 95% within two, whether the curve is narrow or wide. This is the 68–95–99.7 rule. It means the standard deviation works as a built-in ruler for any normal distribution. For a standard deviation of 10, "typical" means between 90 and 110. For a standard deviation of 100, it means between 0 and 200.

L19_OlisBahari_Reflection_ITAI1372_V1.0

The range column also shows the negative-income problem from earlier getting worse. With a standard deviation of 25, only a few values fall below zero. With 100, about 16% of the sample does, since zero is just one standard deviation below the mean.

